

Chapter 16 Depreciation Methods

- **Definition and purpose**

- Depreciation is the reduction in value of an asset over time.
- This reduction is brought on by wear, tear, use, deterioration, obsolescence, etc.
- Depreciation methods give the reduction in value of asset.
- Depreciation amounts are not actual cash flows.
- Depreciation studies are performed for two purposes
 - Book depreciation. This is for accounting purposes.
 - Tax depreciation. This is required to calculate the tax paid to the government on the asset. In most countries, tax depreciation is tax deductible.

- **Depreciation terminology**

- First cost or unadjusted basis, B : Initial purchase price plus all costs incurred for placing the asset in service
- Recovery Period, n : Depreciable life of the asset – often set by law.
- Market Value at time t , MV_t : Amount realized by sale on the open market.
- Salvage Value, S : Estimated trade-in value or market value at the end the asset's useful life

- Depreciation Rate, d_t : The fraction of the first cost removed by depreciation each year.
- Depreciation amount at time t , $D_t = d_t \times B$: Decrease in asset value at time t .
- Book value at time t , $BV_t = B - \sum_{j=0}^t D_j$: Remaining undepreciated capital investment on accounting books at t .
- Personal Property : All property except real estate use in the pursuit of profit or gain
- Real Property : Real estate, buildings and certain structures.

- **Straight line (SL) depreciation**

- Depreciation amount is the same for all years and is given by

$$D_t = (B - S) / n .$$

- That is, depreciation rate is

$$d_t = d = 1/n .$$

- Book value at time t is

$$BV_t = B - tD_t .$$

- **Declining Balance (DB) depreciation**

- The depreciation amount at time t is equal to a fixed percentage of the book value at time $t - 1$,

$$D_t = d \times BV_{t-1} .$$

- Then, D_t and BV_t are given by

$$D_t = d \times B \times (1-d)^{t-1}$$

$$BV_t = B \times (1-d)^t.$$

- The rate d may be estimated based on salvage and book values,

$$d = 1 - (S / B)^{1/n}$$

- In DB, $d \leq 2/n$. If $d = 2/n$, the method name is changed to double declining balance (DDB).

- **Modified accelerated cost recovery system (MACRS)**

- This is the required tax depreciation method in the US. (See publication 946 on the IRS website.)
- The depreciation rate d_t is provided in tabular form.

Year-t	3-Year	5-Year	7-Year	10-Year	15-Year	20-Year
1	0.3333	0.2000	0.1429	0.1000	0.0500	0.0375
2	0.4445	0.3200	0.2449	0.1800	0.0950	0.0722
3	0.1481	0.1920	0.1749	0.1440	0.0855	0.0668
4	0.0741	0.1152	0.1249	0.1152	0.0770	0.0618
5		0.1152	0.0893	0.0922	0.0693	0.0571
6		0.0576	0.0892	0.0737	0.0623	0.0529
7			0.0893	0.0655	0.0590	0.0489
8			0.0446	0.0655	0.0590	0.0452
9				0.0656	0.0591	0.0446
10				0.0655	0.0590	0.0446
11				0.0328	0.0591	0.0446
12					0.0590	0.0446
13					0.0591	0.0446
14					0.0590	0.0446
15					0.0591	0.0446
16					0.0295	0.0446
17						0.0446
18						0.0446
19						0.0446
20						0.0446
21						0.0223
	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

➤ MACRS depreciation rates are a combination of SL, DB, and DDB rates.

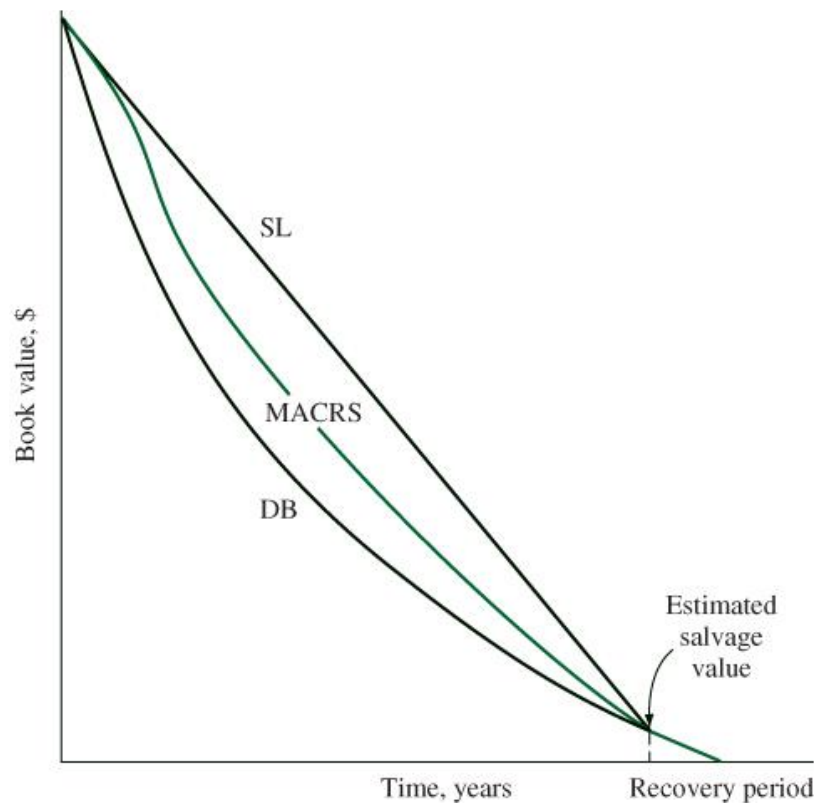
➤ Depreciation amounts and book value are given by $D_t =$

$$d_t \times B \text{ and } BV_t = B - \sum_{j=0}^t D_j .$$

➤ MACRS imposes a built-in *half-year convention*. That is, an asset of life n , is depreciated over “ $n + 1$ ” years.

➤ Half-year convention is based on the assumption that the asset starts service at the middle of year 1 and is placed out of service at the middle of year $n + 1$.

- **Comparing SL, DB, and MACRS**



- **Depletion methods**

- Depreciation is applied to assets that can be replaced.
Depletion applies to resources that are not easily replaced.
E.g., timber, mineral deposits, oil and gas, etc.

- There are two depletion methods: *Cost depletion* and *percentage depletion*.

- Under cost depletion, the annual depletion of a resource is estimated as the year's usage or volume multiplied by a depletion factor p_t , where

$$p_t = (\text{first cost}) / (\text{resource capacity}) .$$

- Under percentage depletion, a constant stated percentage of the resources gross income is depleted each year,¹

$$\text{depletion} = \text{allowable percentage} \times \text{gross income}.$$

- Allowable percentages are as follows:

Deposit	Percentage
Sulfur, uranium, lead, nickel , zinc	22%
Gold, silver ,copper, iron ore and geothermal deposits	15%
Oil and gas wells (varies)	15-22%
Coal, lignite, sodium chloride	10%
Gravel, sand, peat, stone	5%
Most other minerals	14%

¹ This applies as long as the calculated depletion amount does not exceed 50% of the firm's taxable income. However, for oil and gas resources the 50% limit does not apply.